RAMYA SRI MORLA

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EDUCATION

Georgia State University **Aug 2024 - Present**
Master of Science (M.S.), Computer Science *Atlanta, Georgia*

WORK EXPERIENCE

Georgia State University **Aug 2024 - Present**
Research ML Engineer

- Lead an empirical study on job title normalization with large language models, evaluating 8 LLMs using ensemble voting on 10,000 real-world job titles and leveraging NLP and semantic similarity techniques.
- Designed and implemented a scalable experimentation pipeline with semantic clustering (DBSCAN) and batch processing, reducing API calls by 98.9% (80,000 → 7,296) while preserving coverage of 4,560 unique titles.
- Utilized embeddings-based search and clustering to analyze normalization results, exposing fragmentation rates and differences in model behavior across multiple LLM architectures.
- Built analysis and evaluation modules to quantify vocabulary size, fragmentation rates, inter-model agreement (40.7% on average, 0% perfect consensus), and safety-filter slowdowns (8–12×), exposing distinct normalization strategies across model families.
- Developed practical guidelines for production job title normalization systems, including title-only defaults, selective context use for ambiguous titles, and single-model deployments (e.g., Phi-3) that achieve 74–88% better consolidation than multi-LLM ensembles.

Optum **Feb 2023 - Jul 2024**
Junior Software Engineer

- Designed and developed Java-based microservices to support secure, high-volume healthcare transaction processing and real-time data workflows.
- Built and maintained RESTful APIs using Spring Boot, enabling seamless integration with internal healthcare systems and third-party services.
- Integrated Python-based modules for automated ETL and analytics, demonstrating flexibility in tool development.
- Integrated Kafka-based event streaming pipelines for real-time data processing, alerts, and monitoring across backend services.
- Optimized PostgreSQL database queries and batch jobs, improving backend processing performance by approximately 30%.

Rashtriya Ispat Nigam Limited (Vizag Steel Plant) **Apr 2022 - Jun 2022**
Database Engineer Intern

- Developed and optimized 50+ SQL queries, stored procedures, and triggers in Oracle and SQL Server databases, improving data retrieval and reporting speed by 25%.
- Supported database backup, recovery, and disaster recovery drills, validating restore procedures and ensuring 100% data integrity across enterprise systems.
- Collaborated with 5+ cross-functional IT engineers to troubleshoot connectivity issues, fine-tune indexing strategies, and resolve performance bottlenecks, reducing downtime incidents by 30%.
- Documented database administration workflows, optimization techniques, and troubleshooting guides, streamlining onboarding and improving team knowledge-sharing efficiency by 40%.

TECHNICAL SKILLS

- **Programming & Database**: Python, C, C++, Java, Linux/Unix, SQL, MongoDB, Cassandra, MySQL, Oracle, PostgreSQL
- **Frameworks & Tools**: Flask, AWS, Microsoft Azure, Google Cloud Platform, Git, Jenkins, Postman, Eclipse, JIRA
- **Software Engineering**: Object-Oriented Programming (OOP), Data Structures & Algorithms, Debugging, Unit Testing, Code Reviews-, Agile/Scrum
- **AI, ML & NLP**: Machine Learning, NLP, Artificial Intelligence, PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Neural Networks, Embeddings, Semantic Search
- **Search & Data Tools**: Elasticsearch, Data Visualization, Data Cleaning, LLM

PROJECTS

CS Pathways: AI-Powered Course & Career Mapping Tool

- Developed a recommendation system linking university courses to high-demand industry skills using AI-based semantic analysis of syllabi and job postings, improving skill alignment insights for students.
- Designed an interactive website with dashboards and visualizations to display course rankings and skill alignment, enhancing accessibility and user engagement for students.

Diabetes Risk Prediction & Analytics Dashboard

- Performed exploratory data analysis (EDA) and predictive modeling on medical datasets using Logistic Regression, Random Forest, and SVM, identifying key health indicators correlated with diabetes risk.
- Built interactive dashboards in Tableau and Power BI to display patient risk scores, health trends, and feature importance, enabling doctors and hospital staff to track and visualize diabetes risk at scale.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Microsoft Certified: Azure Fundamentals
- Microsoft Certified: Azure Developer Associate